Extended Biography

Moez A. AbdelGawad is an Assistant Professor of Quantum Informatics and Programming Languages at the Informatics Research Institute (IRI), SRTA-City, Egypt. Since 2020, he has also continuously been a remote Visiting Scientist at the Computer Science Department, Rice University (Houston, TX, USA), and an Adjunct Assistant Professor at the Computer & Systems Engineering Department, Faculty of Engineering, Alexandria University (on/off).

The current research of Dr. AbdelGawad focuses on identifying efficient quantum algorithms for classically intractable problems — for example, the graph isomorphism problem and, more generally, the non-commutative hidden subgroup problem (HSP) — using a quantum informatics approach and tools from spectral and algebraic graph theory. To support researchers and students in the field, he developed QAL (Quantum Algorithms Lab), an interactive web application for researching and teaching quantum algorithms and related mathematical topics such as linear algebra and group theory. He is currently documenting the underlying software techniques he developed for QAL and evaluating options for protecting them, including a possible patent filing.

Most recently, Dr. AbdelGawad authored a preprint and presentation introducing QAL, proposed establishing a Quantum Informatics Lab (QI Lab) at IRI, SRTA-City, outlined a vision for how Saudi Arabia might incorporate quantum technologies into hosting the FIFA World Cup 2034, and founded the Quantum Informatics, Algorithms, and Programming (QIAP) online channel.

He has also published on the semantics of generic, nominally-typed object-oriented languages such as Java, C#, Kotlin, C++, and Scala, including a poster at the 2019 Applied Category Theory Conference at Oxford University on a category-theoretic model of generic OOP type systems, along with several further preprints — on induction, coinduction, and fixed points; on partial graph products; on nominal versus structural typing in OOP; and an exploratory piece on self-similar structure in the OOP subtyping relation.

Dr. AbdelGawad earned his Ph.D. in Computer Science from Rice University in 2011, advised by Prof. Robert Cartwright. His dissertation introduced NOOP, a precise mathematical model of nominally-typed object-oriented programming, establishing — for the first time in programming-language research — that inheritance and OO subtyping are effectively identified in mainstream OOP. He adapted the dissertation into a book, published in 2013. Building on that experience, he is also developing an introductory domain theory textbook with Prof. Cartwright and Rebecca Parsons, based on a 2016 monograph draft available on arXiv.

He presented his Ph.D. research at ISDT'13 (International Symposium on Domain Theory, Changsha, China), which was attended by Turing Award laureate Prof. Dana Scott. He was an Adjunct Assistant Professor at the Computer Science Department, Egypt-Japan University of Science & Technology (2014), and a Visiting Scientist at the Category Theory Group, College of Mathematics, Hunan University, China (2015–2016), where he deepened his work on category theory and its applications. In 2005, he interned with Prof. Martin Odersky's Programming Languages Group at EPFL, Switzerland, working on the Scala compiler.

He holds an M.Sc. and B.Sc. in Computer Science from Alexandria University. His M.Sc. thesis introduced 2DEM, a novel two-dimensional encryption mode, presented at SPIE's 2002 conference in Seattle. His B.Sc. graduation project addressed automated human face recognition using neural networks.

Dr. AbdelGawad has a strong, long-standing interest in STEM education — particularly in making mathematics and programming more broadly appealing — using smart devices, games, and puzzles as teaching tools, and including authoring the first Arabic-language instructional videos on solving Rubik's Cube (multipart videos; March 2009). Outside academia, his interests include puzzles and strategy games, diverse reading (such as reading the Qur'an and its various interpretations, history and history of science books, gradually improving his German language skills... etc.), and outdoor activity such as walking, jogging, table tennis, and football/soccer.

Further information on Dr. AbdelGawad's professional activities is available in his CV, available online.